

Section 4.2 - Question 1

Convert the decimal expansion of each of these integers to a binary expansion.

a) 231

- $231 / 2 = 115, r 1$
- $115 / 2 = 57, r 1$
- $57 / 2 = 28, r 1$
- $28 / 2 = 14, r 0$
- $14 / 2 = 7, r 0$
- $7 = 111$

$$231 = 11100111$$

b) 4532

- $4532 / 2 = 2266, r 0$
- $2266 / 2 = 1133, r 0$
- $1133 / 2 = 566, r 1$
- $566 / 2 = 283, r 0$
- $283 / 2 = 141, r 1$
- $141 / 2 = 70, r 1$
- $70 / 2 = 35, r 0$
- $35 / 2 = 17, r 1$
- $17 / 2 = 8, r 1$
- $8 = 1000$

$$4532 = 1000110110100$$

c) 97644

- $97644 / 2 = 48822, r 0, 48822 / 2 = 24411, r 0, 24411 / 2 = 12205, r 1, 12205 / 2 = 6102, r 1, 6102 / 2 = 3051, r 0, 3051 / 2 = 1525, r 1, 1525 / 2 = 762, r 1, 762 / 2 = 381, r 0, 381 / 2 = 190, r 1, 190 / 2 = 95, r 0, 95 / 2 = 47, r 1, 47 / 2 = 23, r 1, 23 / 2 = 11, r 1, 11 / 2 = 5, r 1, 5 / 2 = 2, r 1, 2 / 2 = 1, r 0, 1 / 2 = 0, r 1$

$$97644 = 10111111001101100$$

Section 4.2 - Question 3

Convert the binary expansion of each of these integers to a decimal expansion.

a)

$$(11111)_2$$

$$1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 16 + 8 + 4 + 2 + 1 = 31$$

b)

$$(1000000001)_2$$

$$1 \times 2^9 + 1 \times 2^0 = 512 + 1 = 513$$

c)

$$(101010101)_2$$

$$1 \times 2^8 + 1 \times 2^6 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^0 = 256 + 64 + 16 + 4 + 1 = 341$$

d)

$$(110100100010000)_2$$

$$1 \times 2^{14} + 1 \times 2^{13} + 1 \times 2^{11} + 1 \times 2^8 + 1 \times 2^4 = 16384 + 8192 + 2048 + 256 + 16 = 26896$$

Section 4.2 - Question 5

Convert the octal expansion of each of these integers to a binary expansion.

a)

$$(572)_8$$

$$5 \times 8^2 + 7 \times 8^1 + 2 \times 8^0 = 320 + 56 + 2 = 378$$

b)

$$(1604)_8$$

$$1 \times 8^3 + 6 \times 8^2 + 0 \times 8^1 + 4 \times 8^0 = 512 + 384 + 0 + 4 = 900$$

c)

$$(423)_8$$

$$4 \times 8^2 + 2 \times 8^1 + 3 \times 8^0 = 256 + 16 + 3 = 275$$

d)

$$(2417)_8$$

$$2 \times 8^3 + 4 \times 8^2 + 1 \times 8^1 + 7 \times 8^0 = 1024 + 256 + 8 + 7 = 1295$$

Section 4.2 - Question 6

a)

$$(11110111)_2$$

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 128 + 64 + 32 + 16 + 4 + 2 + 1 = 247$$

$$247/8 = 30, r7 \rightarrow 30/8 = 3, r6 \rightarrow 3/8 = 0, r3$$

$$(367)_8$$

b)

$$(101010101010)_2$$

$$1 \times 2^{11} + 1 \times 2^9 + 1 \times 2^7 + 1 \times 2^5 + 1 \times 2^3 + 1 \times 2^1 = 2048 + 512 + 128 + 32 + 8 + 2 = 2730$$

$$2730/8 = 341, r2 \rightarrow 341/8 = 42, r5 \rightarrow 42/8 = 5, r2 \rightarrow 5/8 = 0, r5$$

$$(5252)_8$$

c)

$$(111011101110111)_2$$

$$1 \times 2^{14} + 1 \times 2^{13} + 1 \times 2^{12} + 1 \times 2^{10} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 16384 + 8192 + 4096 + 1024 + 512 + 256 + 64 + 32 + 16 + 4 + 2 + 1 = 30583$$

$$30583/8 = 3822, r7 \rightarrow 3822/8 = 477, r6 \rightarrow 477/8 = 59, r5 \rightarrow 59/8 = 7, r3 \rightarrow 7/8 = 0, r7$$

$$(73567)_8$$

d)

$$(101010101010101)_2$$

$$1 \times 2^{14} + 1 \times 2^{12} + 1 \times 2^{10} + 1 \times 2^8 + 1 \times 2^6 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^0 = 16384 + 4096 + 1024 + 256 + 64 + 16 + 4 + 1 = 21845$$

$$21845/8 = 2730, r5 \rightarrow 2730/8 = 341, r2 \rightarrow 341/8 = 42, r5 \rightarrow 42/8 = 5, r2 \rightarrow 5/8 = 0, r5$$

$$(52525)_8$$

Section 4.2 - Question 7

a)

$$(80E)_{16}$$

$$8 \times 16^2 + 0 \times 16^1 + 14 \times 16^0 = 2048 + 0 + 14 = 2062$$

$$\begin{aligned} 2062/2 &= 1031, r0 \rightarrow 1031/2 = 515, r1 \rightarrow 515/2 = 257, r1 \rightarrow \\ 257/2 &= 128, r1 \rightarrow 128/2 = 64, r0 \rightarrow 64/2 = 32, r0 \rightarrow 32/2 = \\ 16, r0 \rightarrow 16/2 &= 8, r0 \rightarrow 8/2 = 4, r0 \rightarrow 4/2 = 2, r0 \rightarrow 2/2 = 1, r0 \\ \rightarrow 1/2 &= 0, r1 \end{aligned}$$

$$(100000001110)_2$$

b)

$$(135AB)_{16}$$

$$1 \times 16^4 + 3 \times 16^3 + 5 \times 16^2 + 10 \times 16^1 + 11 \times 16^0 = 65536 + 12288 + 1280 + 160 + 11 = 79275$$

$$(10011010110101011)_2$$

c)

$$(ABBA)_{16}$$

$$10 \times 16^3 + 11 \times 16^2 + 11 \times 16^1 + 10 \times 16^0 = 40960 + 2816 + 176 + 10 = 43962$$

$$(1010101110111010)_2$$

d)

$$(DEFACED)_{16}$$

$$\begin{aligned} 13 \times 16^6 + 14 \times 16^5 + 15 \times 16^4 + 10 \times 16^3 + 12 \times 16^2 + 14 \times 16^1 + \\ 13 \times 16^0 &= 851968 + 917504 + 983040 + 40960 + 3072 + 224 + 13 = \\ 2793781 \end{aligned}$$

$$(1101111011111010110011101101)_2$$

Section 4.2 - Question 8

Convert $(BADFACED)_{16}$ from its hexadecimal expansion to its binary expansion.

$$11 \times 16^7 + 10 \times 16^6 + 13 \times 16^5 + 15 \times 16^4 + 10 \times 16^3 + 12 \times 16^2 + 14 \times 16^1 + 13 \times 16^0 = 3135220973$$

$$\begin{aligned} 3135220973/2 &= 1567610486, r1 \rightarrow 1567610486/2 = 783805243, r0 \\ \rightarrow 783805243/2 &= 391902621, r1 \rightarrow 391902621/2 = 195951310, r1 \\ \rightarrow 195951310/2 &= 97975655, r0 \rightarrow 97975655/2 = 48987827, r1 \\ \rightarrow 48987827/2 &= 24493913, r1 \rightarrow 24493913/2 = 12246956, r1 \\ \rightarrow 12246956/2 &= 6123478, r0 \rightarrow 6123478/2 = 3061739, r0 \\ \rightarrow 3061739/2 &= 1530869, r1 \rightarrow 1530869/2 = 765434, r1 \\ \rightarrow 765434/2 &= 382717, r0 \rightarrow 382717/2 = 191358, r1 \rightarrow \\ 191358/2 &= 95679, r0 \rightarrow 95679/2 = 47839, r1 \rightarrow 47839/2 = \\ 23919, r1 \rightarrow 23919/2 &= 11959, r1 \rightarrow 11959/2 = 5979, r1 \rightarrow \\ 5979/2 &= 2989, r1 \rightarrow 2989/2 = 1494, r1 \rightarrow 1494/2 = 747, r0 \\ \rightarrow 747/2 &= 373, r1 \rightarrow 373/2 = 186, r1 \rightarrow 186/2 = 93, r0 \rightarrow \\ 93/2 &= 46, r1 \rightarrow 46/2 = 23, r0 \rightarrow 23/2 = 11, r1 \rightarrow 11/2 = 5, r1 \\ \rightarrow 5/2 &= 2, r1 \rightarrow 2/2 = 1, r \end{aligned}$$

$$(1011101011011111010110011101101)_2$$

Section 4.2 - Question 23a

Find the sum and product of each of these pairs of numbers. Express your answers as an octal expansion.

$$(763)_8, (147)_8$$

Sum

- $7 + 3 = 10$
 - $\text{digit} = 10 \bmod 8 = 2$
 - $\text{carry} = 10 \div 8 = 1$
- $4 + 6 + 1 = 11$
 - $\text{digit} = 11 \bmod 8 = 3$
 - $\text{carry} = 11 \div 8 = 1$
- $1 + 7 + 1 = 9$
 - $\text{digit} = 9 \bmod 8 = 1$
 - $\text{carry} = 9 \div 8 = 1$

$$(1132)_8$$

Product

- $7 \times 3 = 21$
 - $\text{digit} = 21 \bmod 8 = 5$
 - $\text{carry} = 21 \div 8 = 2$
- $7 \times 6 + 2 = 44$
 - $\text{digit} = 44 \bmod 8 = 4$
 - $\text{carry} = 44 \div 8 = 5$
- $7 \times 7 + 5 = 54$
 - $\text{digit} = 54 \bmod 8 = 6$
 - $\text{carry} = 54 \div 8 = 6$

...Repeat for next two digits of $(147)_8$, shifting one position to the left each time.

$$(144305)_8$$

Section 4.3 - Question 1

Determine whether each of these integers is prime.

a $21 = 3 \times 7$

b 29 is prime.

c 71 is prime.

d 97 is prime.

e $111 = 3 \times 37$

f $143 = 11 \times 13$

Section 4.3 - Question 3

Find the prime factorization of each of these integers.

a $88 = 2^3 \times 11$

b $126 = 2 \times 3^2 \times 7$

c $729 = 3^6$

d $1001 = 7 \times 11 \times 13$

e $1111 = 11 \times 101$

f $909090 = 2 \times 3 \times 5 \times 11 \times 13 \times 37$

Section 4.3 - Question 14

Which positive integers less than 12 are relatively prime to 12?

- $\gcd(1, 12) = 1$
- $\gcd(2, 12) = 2$
- $\gcd(3, 12) = 3$
- $\gcd(4, 12) = 4$
- $\gcd(5, 12) = 1$
- $\gcd(6, 12) = 6$
- $\gcd(7, 12) = 1$
- $\gcd(8, 12) = 4$
- $\gcd(9, 12) = 3$
- $\gcd(10, 12) = 2$
- $\gcd(11, 12) = 1$

1, 5, 7, 11

Section 4.3 - Question 16

a) 21, 34, 55

- $\gcd(21, 34) = 1$
- $\gcd(21, 55) = 1$
- $\gcd(34, 55) = 1$

Yes, they are pairwise relatively prime.

b) 14, 17, 85

- $\gcd(14, 17) = 1$
- $\gcd(14, 85) = 1$
- $\gcd(17, 85) = 17$

Not pairwise relatively prime.

c) 25, 41, 49, 64

- $\gcd(25, 64) = 1$
- $\gcd(41, 49) = 1$
- $\gcd(49, 64) = 1$
- $\gcd(25, 41) = 1$
- $\gcd(41, 64) = 1$
- $\gcd(25, 49) = 1$

Yes, they are pairwise relatively prime.

d) 17, 18, 19, 23

- $\gcd(17, 23) = 1$
- $\gcd(18, 23) = 1$
- $\gcd(19, 23) = 1$
- $\gcd(17, 19) = 1$
- $\gcd(18, 19) = 1$
- $\gcd(17, 18) = 1$

Yes, they are pairwise relatively prime.

Section 4.3 - Question 24

What are the greatest common divisors of these pairs of integers?

a) Pair 1: $(2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2)$

$$\gcd(2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2) = 2700$$

b) Pair 2: $(2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13, 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14})$

$$\gcd(2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13, 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14}) = 66$$

c) Pair 3: $(17, 17^{17})$

$$\gcd(17, 17^{17}) = 17$$

d) Pair 4: $(2^2 \cdot 7, 5^3 \cdot 13)$

$$\gcd(2^2 \cdot 7, 5^3 \cdot 13) = 1$$

e) Pair 5: $(0, 5)$

$$\gcd(0, 5) = 5$$

f) Pair 6: $(2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7)$

$$\gcd(2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7) = 210$$

Section 4.3 - Question 27

What is the least common multiple of each pair in Exercise 25?

a) Pair 1: $(2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2)$

$$\text{lcm}(2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2) = 2^5 \cdot 3^3 \cdot 5^5$$

b) Pair 2: $(2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13, 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14})$

$$\text{lcm}(2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13, 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14}) = 2^{11} \cdot 3^9 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17^{14}$$

c) Pair 3: $(17, 17^{17})$

$$\text{lcm}(17, 17^{17}) = 17^{17}$$

d) Pair 4: $(2^2 \cdot 7, 5^3 \cdot 13)$

$$\text{lcm}(2^2 \cdot 7, 5^3 \cdot 13) = 2^2 \cdot 5^3 \cdot 7 \cdot 13$$

e) Pair 5: $(0, 5)$

$$\text{lcm}(0, 5) = 0$$

f) Pair 6: $(2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7)$

$$\text{lcm}(2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7) = 2 \cdot 3 \cdot 5 \cdot 7$$

Section 4.3 - Question 33

a) $\gcd(12, 18) = 6$

b) $\gcd(111, 201) = 3$

c) $\gcd(1001, 1331) = 11$

d) $\gcd(12345, 54321) = 3$

e) $\gcd(1000, 5040) = 40$

f) $\gcd(9888, 6060) = 12$

Section 4.3 - Question 34

How many divisions are required to find $\gcd(21, 34)$ using the Euclidean algorithm?

1. $34 \% 21 = 13$.
2. $21 \% 13 = 8$.
3. $13 \% 8 = 5$.
4. $8 \% 5 = 3$.
5. $5 \% 3 = 2$.
6. $3 \% 2 = 1$.
7. $2 \% 1 = 0$.

Total divisions required: 7

Section 4.3 - Question 39h

Using the method followed in Example 17, express the greatest common divisor of (3454, 4666) as a linear combination of these integers.

1. $3454 = 0 \cdot 4666 + 3454$
 1. $x = 1 - ((4666//3454) \cdot 0) = 1$
 2. $y = 0 - ((4666//3454) \cdot 1) = -1$
2. $4666 = 1 \cdot 3454 + 1212$
 1. $x = 0 - ((3454//1212) \cdot 1) = -2$
 2. $y = 1 - ((3454//1212) \cdot -1) = 3$
3. $3454 = 2 \cdot 1212 + 1030$
 1. $x = 1 - ((1212//1030) \cdot -2) = 3$
 2. $y = -1 - ((1212//1030) \cdot 3) = -4$
4. $1212 = 1 \cdot 1030 + 182$
 1. $x = -2 - ((1030//182) \cdot 3) = -17$
 2. $y = 3 - ((1030//182) \cdot -4) = 23$
5. $1030 = 5 \cdot 182 + 120$
 1. $x = 3 - ((182//120) \cdot -17) = 20$
 2. $y = -4 - ((182//120) \cdot 23) = -27$
6. $182 = 1 \cdot 120 + 62$
 1. $x = -17 - ((120//62) \cdot 20) = -37$
 2. $y = 23 - ((120//62) \cdot -27) = 50$
7. $120 = 1 \cdot 62 + 58$
 1. $x = 20 - ((62//58) \cdot -37) = 57$
 2. $y = -27 - ((62//58) \cdot 50) = -77$
8. $62 = 1 \cdot 58 + 4$
 1. $x = -37 - ((58//4) \cdot 57) = -835$
 2. $y = 50 - ((58//4) \cdot -77) = 1128$
9. $58 = 14 \cdot 4 + 2$
 1. $x = 57 - ((4//2) \cdot -835) = 1727$
 2. $y = -77 - ((4//2) \cdot 1128) = -2333$
10. $4 = 2 \cdot 2 + 0$

GCD: 2

Linear combination: $-835 \cdot 4666 + 1128 \cdot 3454 = 2$

Section 4.3 - Question 42

Use the extended Euclidean algorithm to express $\gcd(252, 356)$ as a linear combination of 252 and 356.

1. $252 = 0 \cdot 356 + 252$
 1. $x = 1 - ((356//252) \cdot 0) = 1$
 2. $y = 0 - ((356//252) \cdot 1) = -1$
2. $356 = 1 \cdot 252 + 104$
 1. $x = 0 - ((252//104) \cdot 1) = -2$
 2. $y = 1 - ((252//104) \cdot -1) = 3$
3. $252 = 2 \cdot 104 + 44$
 1. $x = 1 - ((104//44) \cdot -2) = 5$
 2. $y = -1 - ((104//44) \cdot 3) = -7$
4. $104 = 2 \cdot 44 + 16$
 1. $x = -2 - ((44//16) \cdot 5) = -12$
 2. $y = 3 - ((44//16) \cdot -7) = 17$
5. $44 = 2 \cdot 16 + 12$
 1. $x = 5 - ((16//12) \cdot -12) = 17$
 2. $y = -7 - ((16//12) \cdot 17) = -24$
6. $16 = 1 \cdot 12 + 4$
 1. $x = -12 - ((12//4) \cdot 17) = -63$
 2. $y = 17 - ((12//4) \cdot -24) = 89$
7. $12 = 3 \cdot 4 + 0$

GCD: 4

Linear combination: $17 \cdot 356 + -24 \cdot 252 = 4$